

Methods of Gently Notifying Pedestrians of Approaching Objects when Listening to Music

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ABSTRACT

Many people now listen to music with earphones while walking, and are less likely to notice approaching people, cars, etc. Many methods of detecting approaching objects and notifying pedestrians have been proposed, but few have focused on low urgency situations or music listeners, and many notification methods are unpleasant. Therefore, in this work, we propose methods of gently notifying pedestrians listening to music of approaching objects using environmental sound. We conducted experiments in a virtual environment to assess directional perception accuracy and comfort. Our results show the proposed method allows participants to detect the direction of approaching objects as accurately as explicit notification methods, with less discomfort.

CCS CONCEPTS

• Human-centered computing → Auditory feedback.

KEYWORDS

Auditory Display, Directional Information Presentation, Sound Localization, Environmental Sound

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1 INTRODUCTION

More people are listening to music using earphones or headphones while walking, reducing awareness of their surroundings [12]. Although functions that capture external sounds (e.g., ambient sound

mode [8]) are widely used to address this problem, it remains difficult to perceive the approach of people, bicycles, or automobiles when listening to music, resulting in obstruction of or collisions with, other users of roads and sidewalks.

Many methods have been proposed for detecting surrounding objects and notifying pedestrians of their approach. However, some are unsuitable for detecting quietly approaching objects [13], while others require additional devices for notification by tactile stimuli [2, 6]. Many are designed for urgent situations and use jarring notification sounds like beeps [5], which may be annoying in non-emergency situations. Furthermore, the accuracy of directional perception when using less intrusive sounds during music listening has not been evaluated. Some methods tweak the music itself, affecting the music listening experience [3, 4, 10].

Our end goal is to develop a device that allows pedestrians to enjoy music to the maximum, safely and without disturbing other traffic. To achieve this goal, we propose detecting approaching objects and varying the notification method according to the situation, giving gentler notifications in less urgent situations. In this poster, as the first step, we proposed methods to present environmental sounds from only the direction of an object approaching with environmental sounds blocked and made the following contributions.

- Verification of the directional perception accuracy of several environmental sound notification methods during music listening, which has not been evaluated.
- Comparison of subjective evaluations between the proposed and explicit notification methods for elements that could be perceived as unpleasant when walking during music listening.

We synthesized these results and investigated a method of notification that balances less discomfort and accuracy in perceiving the direction of approaching objects during music listening.

2 EXPERIMENT

We conducted the experiments to verify the appropriateness of the proposed methods for gently warning pedestrians.

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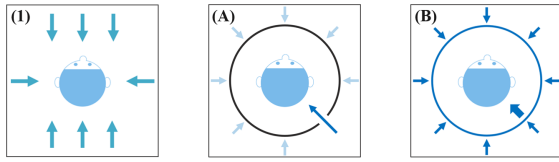


Figure 1: (1) Eight directions of approaching objects. (A) (B) Sound presentation methods.

2.1 Experimental Configuration

We conducted our experiment in a virtual environment because conditions are easier to control than in real-world environments. We used a 360-degree camera (RICOH THETA S) and an ambisonic microphone (Sennheiser AMBEO VR MIC) to capture video and record audio at fixed points in multiple environments (quiet places, beside intersections, etc.). Each scene involved video of a person or car approaching from one of eight directions, with multiple scenes for each direction. The eight directions in this work were the approaching directions that could occur when walking, as shown in Figure 1. We processed recorded audio with various plug-ins, placed these sound sources in eight locations around the participants in Unity, and presented the spatial sound using the Steam Audio SDK [9]. To reproduce the sound field of actual environments, we played audio recorded with eight omnidirectional microphones back through speakers placed at eight locations in a circle 1 meter from the participants.

We compared six notification methods: *Beeps* (an intermittent sound at 1000 Hz), *Voice* (a repetitive "Approaching" using Google UK English Female, a method said to be less harsh than *Beeps* [1]), and four methods of presenting environmental sound (methods (A) and (B) shown in Figure 1, at two different sound pressure levels). (A) is to present only sounds in the direction of an object approaching. While this can preserve an object's actual direction of arrival, it requires special microphones that can collect and reproduce spatial sound [7]. Also, objects approaching from directions producing little sound can be difficult to perceive [11]. (B) is to present all environmental sounds as if they were originating from the direction of the approaching object. This has the advantage of not requiring a special microphone, but it manipulates the direction of ambient sound arrival, which may cause discomfort for some users.

The sound pressure level (SPL) for each environmental sound notification method was measured by inserting an omnidirectional microphone into a dummy head wearing headphones to reproduce headphone listening. *Beeps*, *Voice*, and the music played in each scene were all measured at around 55dB. The lower SPL for environmental sound notification was equivalent to that measured in the actual environment (around 45dB), while the higher SPL, denoted as "-Adjusted", was presented at the same level as the music.

2.2 Experimental Procedure

Participants wore an Head Mounted Display (HMD) (HTC Vive) and noise-canceling headphones (Sony WH-1000MX3), and held a controller (SIE DUALSHOCK 4) in their hands. The HMD played the VR video of objects approaching, and we played the sound from the surrounding speakers to match the video. We presented one

type of notification sound in one trial. We prepared several sets of scenes by randomly combining scenes so that the number of approaches from each direction was the same and used different sets in each trial. The order of each trial was randomized.

Participants used the controller to perform two tasks simultaneously during each scene. Task 1: When participants heard notification sounds of approaching objects, they were to move the control stick in the direction of the sounds (referring to the video and the change in sound caused by the head direction) and press a button as quickly as possible. Task 2: When participants sensed a change in the music playback speed, they pressed a button. This was a distraction task so participants would not concentrate too much on perceiving notification sounds, since pedestrians do not always walk with the sound in mind. We used some music that most participants would have encountered. At the end of each trial, participants completed a 7-level semantic differential scale questionnaire regarding the notification sounds.

We evaluated the participants' directional perception accuracy, time from notification to direction designation, and Task 2 accuracy. We also tabulated questionnaire responses for each notification method regarding "interference with music", "discomfort with the sound itself", "sense of urgency", "unnaturalness for visual information", and "concentration required to perceive the direction of notification". Twelve males and females aged 22 to 49 with no visual or auditory impairments participated in the experiment. Eight participants said they regularly listened to music while walking.

2.3 Result and Discussion

As shown in Figure 2, the participants generally perceived the direction of *Beeps* and *Voice* accurately, and the number of times they failed to hear the notification was negligible. Environmental sound (A)-Adjusted and (B)-Adjusted, participants perceived the location of the notification as accurately as they did with *Beeps* and *Voice*, although failures to hear the notification increased slightly. In contrast, when using (A) and (B), participants often misidentified the direction of the notifications and often failed to hear the notifications. This suggests that the SPL of the notification sound should be equal to or higher than that of the user's music if users are to perceive the direction of notifications accurately. (A) and (B) are not mentioned below. We obtained no notable results regarding time from notification to direction indication or Task 2 accuracy.

The subjective evaluation of the notification sound showed a significant difference only between *Beeps* and (A)-Adjusted in "sense of urgency" shown in Figure 3 due to the small number of participants (Wilcoxon signed-rank test with Holm correction, $p=0.045 < 0.05$). However, a trend toward lower (better) ratings for both (A)-Adjusted and (B)-Adjusted was observed, except for "concentration required to perceive direction", compared to *Beeps* and *Voice*. Interestingly, several participants who frequently listen to music while walking, target of this work, said they did not find the use of environmental sound notifications uncomfortable, while some non-target participants found it uncomfortable.

Our directional perception accuracy and subjective evaluation results suggest that (A)-Adjusted and (B)-Adjusted are likely to be suitable notification methods for non-emergency situations. Contrary to expectations, there was little difference in "unnaturalness

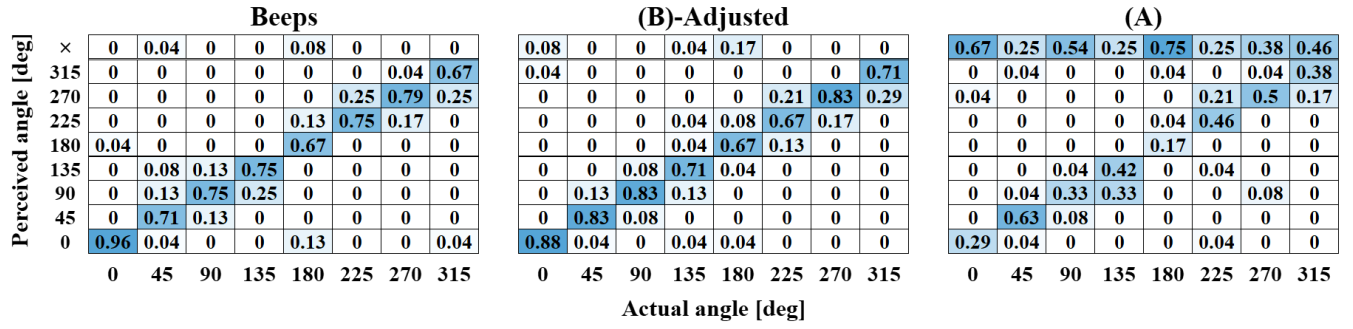


Figure 2: Directional perception accuracy results. Angles are clockwise angles with the front face at 0 degrees. Numerical values indicate the percentage of times participants responded that it was the angle of that row among the number of times they were presented with the angle of that column. "x" responses for "Actual angle" indicates participants did not hear the notification sound and did not respond.

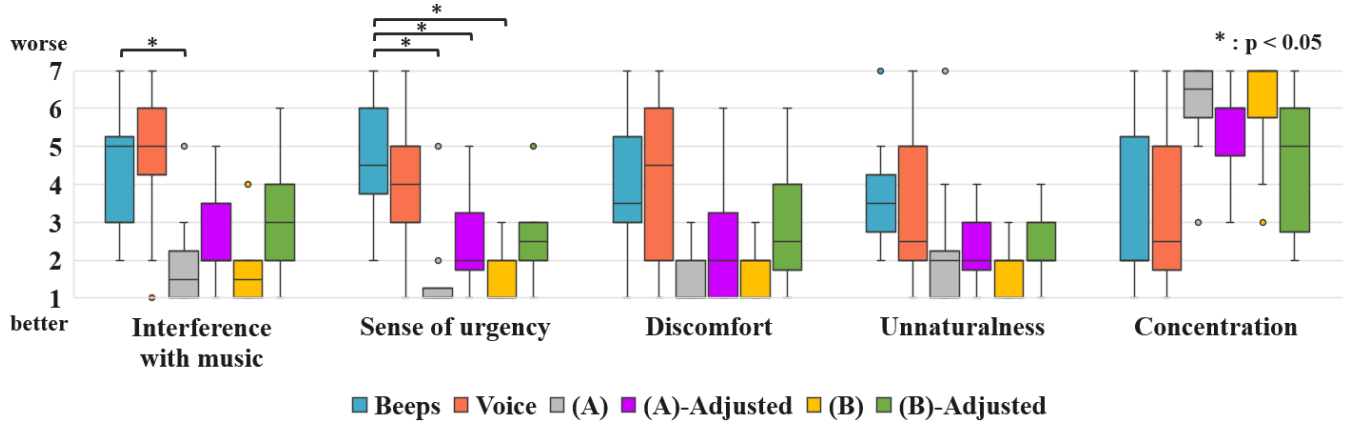


Figure 3: Subjective evaluations of the notification sounds.

for visual information" between the two environmental sound notification methods, as shown in Figure 3. We assume this was because it was difficult to hear the details of the notification sounds when listening to music, and the sound's actual arrival direction was not a concern. However, the ratings other than unnaturalness and concentration were slightly higher (worse) for (B)-Adjusted than for (A)-Adjusted. This may be attributed to the amount of information included in the notification sound. Although there may be a trade-off between user comfort and ease of use, considering that (B)-Adjusted does not require a special microphone, this rating difference is acceptable, and (B)-Adjusted may be used in future work. On the other hand, the environmental sound notification methods tended to require more user concentration to perceive the direction of notification than *Beeps* or *Voice*, so it may be necessary to consider ways to mitigate this problem.

3 FUTURE WORK

We need to conduct experiments in real-world environments due to the differences between simulated and real visual and acoustic information. After determining the most suitable methods for

notifying pedestrians of approaching objects in routine and emergency situations, we will propose and evaluate a device that allows pedestrians to enjoy music without disrupting traffic.

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